Updated announcement as of 23 Oct. 2026

Reading Materials: pages 10-18 out of "ENGINEERING DESIGN", by George E . Dieter & Linda C . Schmidt, McGraw-hill. *Pages are given by the end of this package.*

RULES

1- **Time/date**: the first ten minutes of Thursday 17 Dec Lecture
2- **Location:** GE211- regular-class room
3- **Pass grade** starts from **60%**
4- **If passed**, one defect from the concept session will be removed
5- **No penalties** by any mean upon not attending and/or passing the quiz
6- **Resubmission**: There is no resubmission of the DR-quiz
7- **Grades**: The DR-grades will be included in the course final grade but will not be announces separately
8- **Checklists**: use the attached checklist only.
9- **Quiz paper** will not be returned to the student
10- **Wish** you all good luck

Review the course link

Qassim University **GE-211** **College of Engineering**

481 DR-Quiz semester

Name: ..... **Quiz Date:**

Group Number | **Team Number** | **Member Number**

Quiz Reading Assignment: pages 10-18 out of "ENGINEERING DESIGN", by G.Dieter & L.Schmidt

Work Evaluated by ..... Assessment Symbol/Color ..... Date

You are to complete **only the top 2 lines** of this page. The grader will complete this checklist

Yes	No	Self-Regulation Issues
		A. The quiz was submitted on time (if No, instructor initial under No)
		B. This checklist is on the top of the quiz-sheet and with the correct orientation
		C. This checklist is stapled from the left to the quiz-sheet
		D. **Only** the top 2 lines of this page are completed by the student prior to the submission and correct.
		E. There were no other Self-Regulation Problems (describe if No _____)
		Enter a 1 (one) in the cell to left if any of the items listed above are marked as **No**

%	Summary of Quiz Performances
	Enter the % of quiz questions correctly answered in the box to the left

RESULTS OF WORK ASSESSMENT

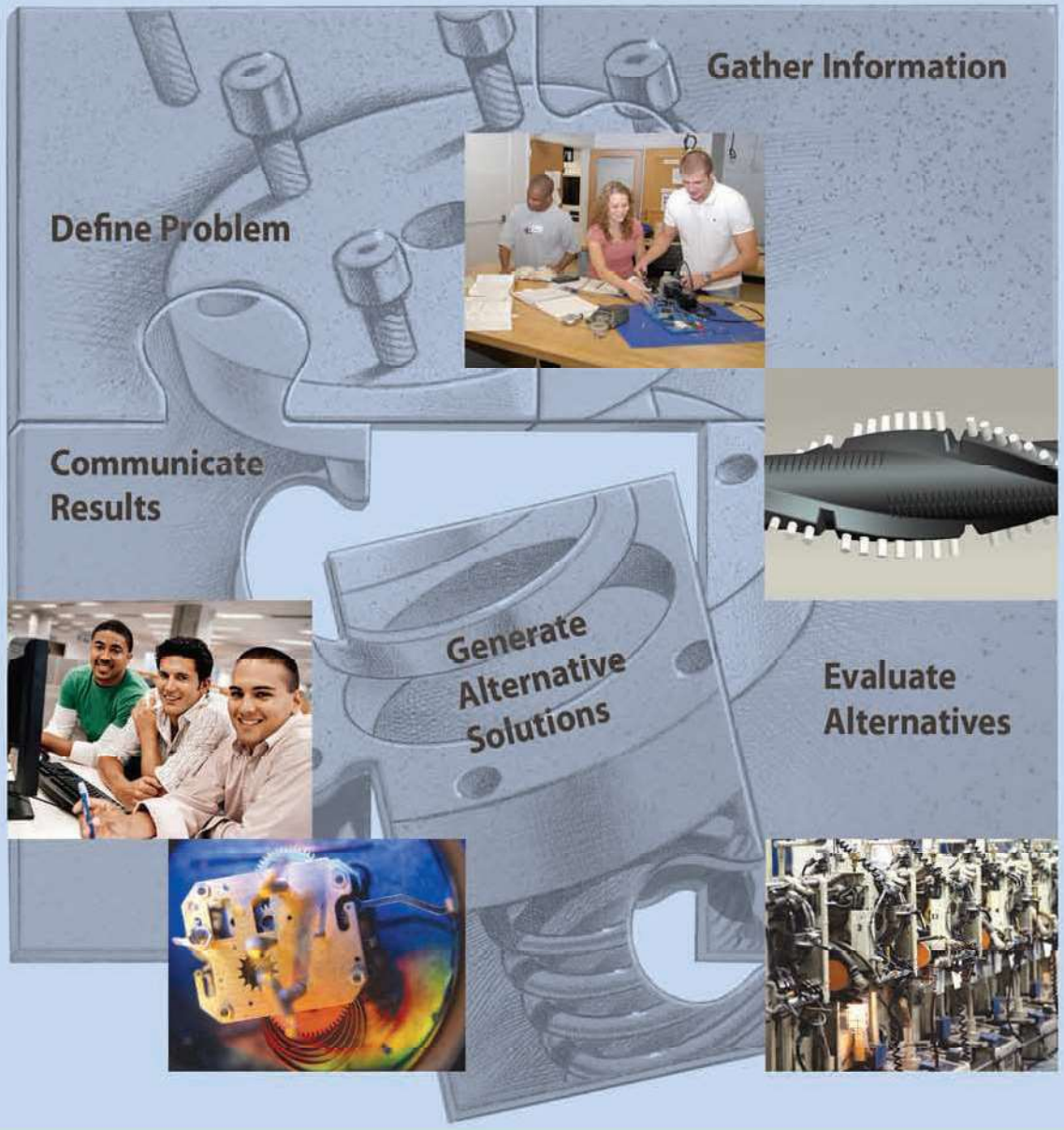
Grade	
Pass	Greater than or equal 60% without any No
Not pass	otherwise

1- **If passed**, one defect from the concept session will be removed
2- **No penalties** by any mean upon not attending, passing the quiz
3- **Resubmission**: There is no resubmission of the DR-quiz
4- **Grades**: The DR-grades will be included in the course final grade but will not be announces
5- Quiz paper will not be returned to the student

Wish you all good luck

Engineering Design

Fourth Edition



George E. Dieter • Linda C. Schmidt

ENGINEERING DESIGN

FOURTH EDITION

George E. Dieter
University of Maryland

Linda C. Schmidt
University of Maryland

McGraw-Hill Higher Education

Boston Burr Ridge, IL Dubuque, IA New York San Francisco St. Louis
Bangkok Bogotá Caracas Kuala Lumpur Lisbon London Madrid Mexico City
Milan Montreal New Delhi Santiago Seoul Singapore Sydney Taipei Toronto

DETAILED CONTENTS

Preface xxiii

Chapter 1 Engineering Design **1**

1.1 Introduction 1
1.2 Engineering Design Process 3
1.2.1 Importance of the Engineering Design Process 4
1.2.2 Types of Designs 5
1.3 Ways to Think About the Engineering Design Process 6
1.3.1 A Simplified Iteration Model 6
1.3.2 Design Method Versus Scientific Method 8
1.3.3 A Problem-Solving Methodology 10
1.4 Considerations of a Good Design 14
1.4.1 Achievement of Performance Requirements 14
1.4.2 Total Life Cycle 17
1.4.3 Regulatory and Social Issues 18
1.5 Description of Design Process 19
1.5.1 Phase I. Conceptual Design 19
1.5.2 Phase II. Embodiment Design 20
1.5.3 Phase III. Detail Design 21
1.5.4 Phase IV. Planning for Manufacture 22
1.5.5 Phase V. Planning for Distribution 23
1.5.6 Phase VI. Planning for Use 23
1.5.7 Phase VII. Planning for Retirement of the Product 23
1.6 Computer-Aided Engineering 24
1.7 Designing to Codes and Standards 26
1.8 Design Review 29
1.8.1 Redesign 30
1.9 Societal Considerations in Engineering Design 31

1.3.3 A Problem-Solving Methodology

Designing can be approached as a problem to be solved. A problem-solving methodology that is useful in design consists of the following steps.[8]

- Definition of the problem
- Gathering of information
- Generation of alternative solutions
- Evaluation of alternatives and decision making
- Communication of the results

This problem-solving method can be used at any point in the design process, whether at the conception of a product or the design of a component.

Definition of the Problem

The most critical step in the solution of a problem is the *problem definition* or formulation. The true problem is not always what it seems at first glance. Because this step seemingly requires such a small part of the total time to reach a solution, its importance is often overlooked. Figure 1.4 illustrates how the final design can differ greatly depending upon how the problem is defined.

The formulation of the problem should start by writing down a problem statement. This document should express as specifically as possible what the problem is. It should include objectives and goals, the current state of affairs and the desired state, any constraints placed on solution of the problem, and the definition of any special technical terms. The problem-definition step in a design project is covered in detail in Chap. 3.

Problem definition often is called *needs analysis.* While it is important to identify the needs clearly at the beginning of a design process, it should be understood that this is difficult to do for all but the most routine design. It is the nature of the design process that new needs are established as the design process proceeds because new problems arise as the design evolves. At this point, the analogy of design as problem solving is less fitting. Design is problem solving only when all needs and potential issues with alternatives are known. Of course, if these additional needs require reworking those parts of the design that have been completed, then penalties are incurred

8. A similar process called the guided iteration methodology has been proposed by J. R. Dixon; see J. R. Dixon and C. Poli, *Engineering Design and Design for Manufacturing,* Field Stone Publishers, Conway, MA, 1995. A different but very similar problem-solving approach using TQM tools is given in Sec. 4.7.

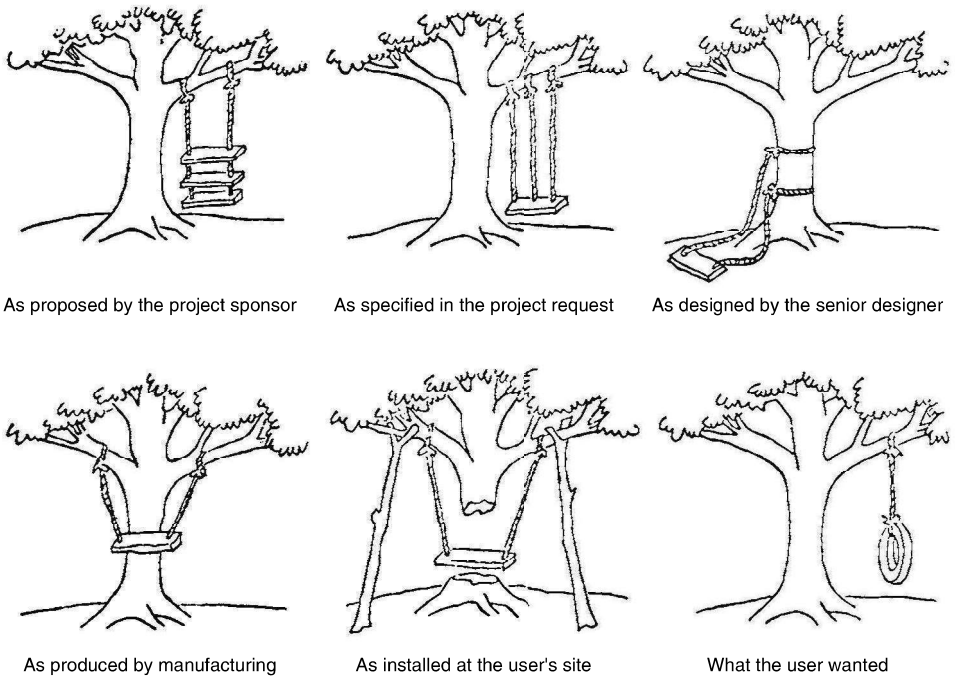


FIGURE 1.4
Note how the design depends on the viewpoint of the individual who defines the problem.

in terms of cost and project schedule. Experience is one of the best remedies for this aspect of designing, but modern computer-based design tools help ameliorate the effects of inexperience.

Gathering Information

Perhaps the greatest frustration you will encounter when you embark on your first design project will be either the dearth or the plethora of information. No longer will your responsibility stop with the knowledge contained in a few chapters of a text. Your assigned problem may be in a technical area in which you have no previous background, and you may not have even a single basic reference on the subject. At the other extreme you may be presented with a mountain of reports of previous work, and your task will be to keep from drowning in paper. Whatever the situation, the immediate task is to identify the needed pieces of information and find or develop that information.

An important point to realize is that the information needed in design is different from that usually associated with an academic course. Textbooks and articles published in the scholarly technical journals usually are of lesser importance. The need often is for more specific and current information than is provided by those sources. Technical reports published as a result of government-sponsored R&D, company reports, trade journals, patents, catalogs, and handbooks and literature published by

vendors and suppliers of material and equipment are important sources of information. The Internet is becoming a very useful resource. Often the missing piece of information can be supplied by an Internet search, or by a telephone call or an e-mail to a key supplier. Discussions with in-house experts (often in the corporate R&D center) and outside consultants may prove helpful.

The following are some of the questions concerned with obtaining information:

What do I need to find out?
Where can I find it and how can I get it?
How credible and accurate is the information?
How should the information be interpreted for my specific need?
When do I have enough information?
What decisions result from the information?

The topic of information gathering is discussed in Chap. 5.

Generation of Alternative Solutions

Generating alternative solutions or design concepts involves the use of creativity-stimulation methods, the application of physical principles and qualitative reasoning, and the ability to find and use information. Of course, experience helps greatly in this task. The ability to generate high-quality alternative solutions is vital to a successful design. This important subject is covered in Chap. 6, Concept Generation.

Evaluation of Alternatives and Decision Making

The evaluation of alternatives involves systematic methods for selecting the best among several concepts, often in the face of incomplete information. Engineering analysis procedures provide the basis for making decisions about service performance. Design for manufacturing analyses (Chap. 13) and cost estimation (Chap. 16) provide other important information. Various other types of engineering analysis also provide information. Simulation of performance with computer models is finding wide usage (Chap. 10). Simulated service testing of an experimental model and testing of full-sized prototypes often provide critical data. Without this quantitative information it is not possible to make valid evaluations.

Several methods for evaluating design concepts, or any other problem solution, are given in Chap. 7.

An important activity at every step in the design process, but especially as the design nears completion, is *checking*. In general, there are two types of checks that can be made: mathematical checks and engineering-sense checks. Mathematical checks are concerned with checking the arithmetic and the equations for errors in the conversion of units used in the analytical model. Incidentally, the frequency of careless math errors is a good reason why you should adopt the practice of making all your design calculations in a bound notebook. In that way you won't be missing a vital calculation when you are forced by an error to go back and check things out. Just draw a line through the section in error and continue. It is of special importance to ensure that every equation is dimensionally consistent.

Engineering-sense checks have to do with whether the answers "seem right." Even though the reliability of your intuition increases with experience, you can now develop the habit of staring at your answer for a full minute, rather than rushing on to do the

next calculation. If the calculated stress is 10^6 psi, you know something went wrong! Limit checks are a good form of engineering-sense check. Let a critical parameter in your design approach some limit (zero, infinity, etc.), and observe whether the equation behaves properly.

We have stressed the *iterative* nature of design. An optimization technique aimed at producing a *robust design* that is resistant to environmental influences (water vapor, temperature, vibration, etc.) most likely will be employed to select the best values of key design parameters (see Chap. 15).

Communication of the Results

It must always be kept in mind that the purpose of the design is to satisfy the needs of a customer or client. Therefore, the finalized design must be properly communicated, or it may lose much of its impact or significance. The communication is usually by oral presentation to the sponsor as well as by a written design report. Surveys typically show that design engineers spend 60 percent of their time in discussing designs and preparing written documentation of designs, while only 40 percent of the time is spent in analyzing and testing designs and doing the designing. Detailed engineering drawings, computer programs, 3-D computer models, and working models are frequently among the "deliverables" to the customer.

It hardly needs to be emphasized that communication is not a one-time occurrence to be carried out at the end of the project. In a well-run design project there is continual oral and written dialog between the project manager and the customer. This extremely important subject is considered in greater depth in Chap. 9.

Note that the problem-solving methodology does not necessarily proceed in the order just listed. While it is important to define the problem early on, the understanding of the problem improves as the team moves into solution generation and evaluation. In fact, design is characterized by its iterative nature, moving back and forth between partial solutions and problem definition. This is in marked contrast with engineering analysis, which usually moves in a steady progression from problem setup to solution.

There is a paradox inherent in the design process between the accumulation of problem (domain) knowledge and freedom to improve the design. When one is creating an original design, very little is known about its solution. As the design team proceeds with its work; it acquires more knowledge about the technologies involved and the possible solutions (Fig. 1.5). The team has moved up the learning curve. However, as the design process proceeds, the design team is forced to make many decisions about design details, technology approaches, perhaps to let contracts for long-lead-time equipment, and so on. Thus, as Fig. 1.5 shows, the freedom of the team to go back and start over with their newly gained knowledge (experience) decreases greatly as their knowledge about the design problem grows. At the beginning the designer has the freedom to make changes without great cost penalty, but may not know what to do to make the design better. The paradox comes from the fact that when the design team finally masters the problem, their design is essentially frozen because of the great penalties involved with a change. The solution is for the design team to learn as much about the problem as early in the design process as it possibly can. This also places high priority on the team members learning to work independently toward a common goal (Chap. 4), being skilled in gathering information (Chap. 5), and being

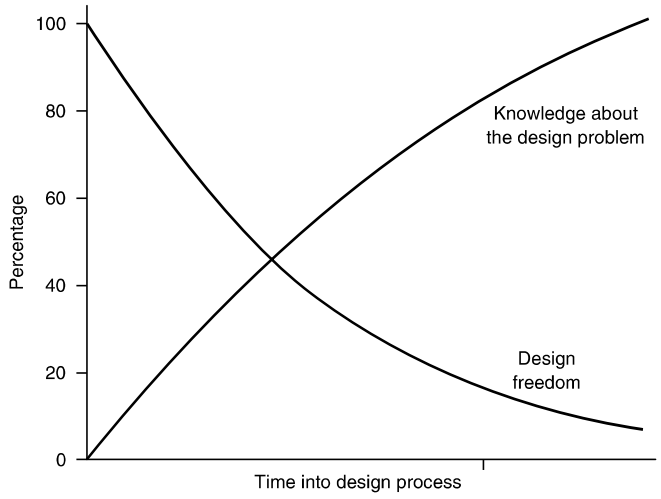


FIGURE 1.5
The design paradox between design knowledge and design freedom.

good at communicating relevant knowledge to their teammates. Design team members must become stewards of the knowledge they acquire. Figure 1.5 also shows why it is important to document in detail what has been done, so that the experience can be used by subsequent teams in future projects.

1.4 CONSIDERATIONS OF A GOOD DESIGN

Design is a multifaceted process. To gain a broader understanding of engineering design, we group various considerations of good design into three categories: (1) achievement of performance requirements, (2) life-cycle issues, and (3) social and regulatory issues.

1.4.1 Achievement of Performance Requirements

It is obvious that to be feasible the design must demonstrate the required performance. Performance measures both the function and the behavior of the design, that is, how well the device does what it is designed to do. Performance requirements can be divided into primary performance requirements and complementary performance requirements. A major element of a design is its *function*. The function of a design is how it is expected to behave. For example, the design may be required to grasp an object of a certain mass and move it 50 feet in one minute. Functional requirements are usually expressed in capacity measures such as forces, strength, deflection, or energy or power output or consumption. Complementary performance requirements are concerns such as the useful life of the design, its robustness to factors occurring in the

service environment (see Chap. 15), its reliability (see Chap. 14), and ease, economy, and safety of maintenance. Issues such as built-in safety features and the noise level in operation must be considered. Finally, the design must conform to all legal requirements and design codes.

A product is usually made up of a collection of parts, sometimes called piece-parts. A *part* is a single piece requiring no assembly. When two or more parts are joined it is called an *assembly.* Often large assemblies are composed of a collection of smaller assemblies called *subassemblies.* A similar term for part is *component.* The two terms are used interchangeably in this book, but in the design literature the word *component* sometimes is used to describe a subassembly with a small number of parts. Consider an ordinary ball bearing. It consists of an outer ring, inner ring, 10 or more balls depending on size, and a retainer to keep the balls from rubbing together. A ball bearing is often called a component, even though it consists of a number of parts.

Closely related to the function of a component in a design is its form. *Form* is what the component looks like, and encompasses its shape, size, and surface finish. These, in turn, depend upon the material it is made from and the manufacturing processes that are used to make it.

A variety of analysis techniques must be employed in arriving at the features of a component in the design. By *feature* we mean specific physical attributes, such as the fine details of geometry, dimensions, and tolerances on the dimensions.[9] Typical geometrical features would be fillets, holes, walls, and ribs. The computer has had a major impact in this area by providing powerful analytical tools based on finite-element analysis. Calculations of stress, temperature, and other field-dependent variables can be made rather handily for complex geometry and loading conditions. When these analytical methods are coupled with interactive computer graphics, we have the exciting capability known as computer-aided engineering (CAE); see Sec. 1.6. Note that with this enhanced capability for analysis comes greater responsibility for providing better understanding of product performance at early stages of the design process.

Environmental requirements for performance deal with two separate aspects. The first concerns the service conditions under which the product must operate. The extremes of temperature, humidity, corrosive conditions, dirt, vibration, and noise, must be predicted and allowed for in the design. The second aspect of environmental requirements pertains to how the product will behave with regard to maintaining a safe and clean environment, that is, green design. Often governmental regulations force these considerations in design, but over time they become standard design practice. Among these issues is the disposal of the product when it reaches its useful life. For more information on design for environment (DFE), see Sec. 8.9.

Aesthetic requirements refer to "the sense of the beautiful." They are concerned with how the product is perceived by a customer because of its shape, color, surface texture, and also such factors as balance, unity, and interest. This aspect of design usually is the responsibility of the industrial designer, as opposed to the engineering designer. The industrial designer is an applied artist. Decisions about the appearance of the product should be an integral part of the initial design concept. An important

9. In product development the term *feature* has an entirely different meaning as "an aspect or characteristic of the product." For example, a product feature for a power drill could be a laser beam attachment for alignment of the drill when drilling a hole.

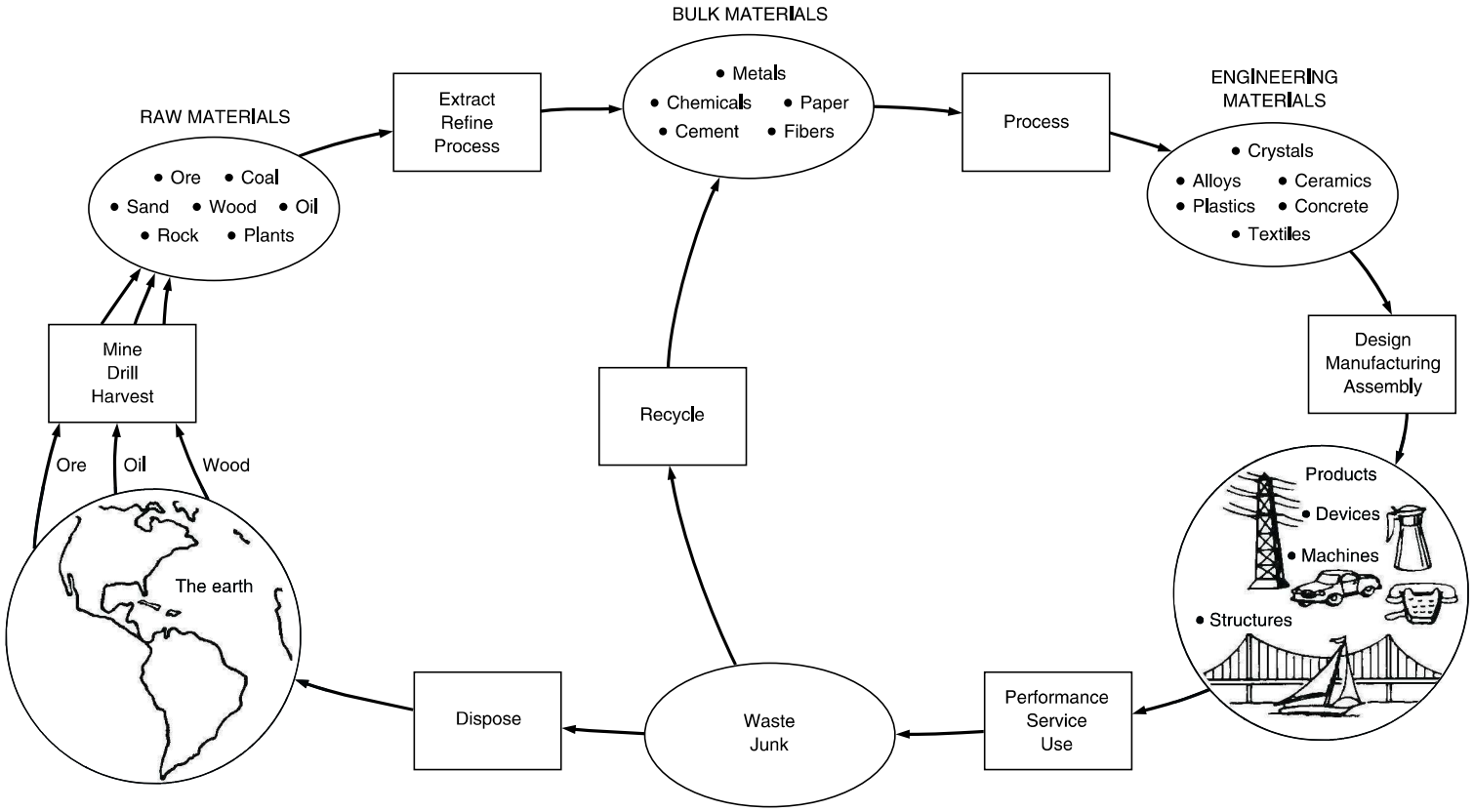


FIGURE 1.6
The total materials cycle. (*Reproduced from "Materials and Man's Needs," National Academy of Sciences, Washington, D.C., 1974.*)

design consideration is adequate attention to *human factors engineering,* which uses the sciences of biomechanics, ergonomics, and engineering psychology to assure that the design can be operated efficiently by humans. It applies physiological and anthropometric data to such design features as visual and auditory display of instruments and control systems. It is also concerned with human muscle power and response times. The industrial designer often is responsible for considering the human factors. For further information, see Sec. 8.8.

Manufacturing technology must be closely integrated with product design. There may be restrictions on the manufacturing processes that can be used, because of either selection of material or availability of equipment within the company.

The final major design requirement is cost. Every design has requirements of an economic nature. These include such issues as product development cost, initial product cost, life cycle product cost, tooling cost, and return on investment. In many cases cost is the most important design requirement. If preliminary estimates of product cost look unfavorable, the design project may never be initiated. Cost enters into every aspect of the design process. Procedures for estimating costs are considered in Chap. 16 and the subject of economic decision making (engineering economics) is presented in Chap. 18.[10]

1.4.2 Total Life Cycle

The total life cycle of a part starts with the conception of a need and ends with the retirement and disposal of the product.

Material selection is a key element in shaping the total life cycle (see Chap. 11). In selecting materials for a given application, the first step is evaluation of the service conditions. Next, the properties of materials that relate most directly to the service requirements must be determined. Except in almost trivial conditions, there is never a simple relation between service performance and material properties. The design may start with the consideration of static yield strength, but properties that are more difficult to evaluate, such as fatigue, creep, toughness, ductility, and corrosion resistance may have to be considered. We need to know whether the material is stable under the environmental conditions. Does the microstructure change with temperature and therefore change the properties? Does the material corrode slowly or wear at an unacceptable rate?

Material selection cannot be separated from *manufacturability* (see Chap. 13). There is an intimate connection between design and material selection and the manufacturing processes. The objective in this area is a trade-off between the opposing factors of minimum cost and maximum durability. Durability is the amount of use one gets from a product before it is no longer useable. Current societal issues of energy conservation, material conservation, and protection of the environment result in new pressures in the selection of materials and manufacturing processes. Energy costs, once nearly ignored in design, are now among the most prominent design considerations. Design for materials recycling also is becoming an important consideration.

The life cycle of production and consumption that is characteristic of all products is illustrated by the materials cycle shown in Fig. 1.6. This starts with the mining of a

10. Chapter 18 can be found on the website for this text, www.mhhe.com/dieter.

mineral or the drilling for oil or the harvesting of an agricultural fiber such as cotton. These raw materials must be processed to extract or refine a bulk material (e.g., an aluminum ingot) that is further processed into a finished engineering material (e.g., an aluminum sheet). At this stage an engineer designs a product that is manufactured from the material, and the part is put into service. Eventually the part wears out or becomes obsolete because a better product comes on the market. At this stage, one option is to junk the part and dispose of it in some way that eventually returns the material to the earth. However, society is becoming increasingly concerned with the depletion of natural resources and the haphazard disposal of solid materials. Thus, we look for economical ways to recycle waste materials (e.g., aluminum beverage cans).

1.4.3 Regulatory and Social Issues

Specifications and standards have an important influence on design practice. The standards produced by such societies as ASTM and ASME represent voluntary agreement among many elements (users and producers) of industry. As such, they often represent minimum or least-common-denominator standards. When good design requires more than that, it may be necessary to develop your own company or agency standards. On the other hand, because of the general nature of most standards, a standard sometimes requires a producer to meet a requirement that is not essential to the particular function of the design.

The codes of ethics of all professional engineering societies require the engineer to protect public health and safety. Increasingly, legislation has been passed to require federal agencies to regulate many aspects of safety and health. The requirements of the Occupational Safety and Health Administration (OSHA), the Consumer Product Safety Commission (CPSC), the Environmental Protection Agency (EPA), and the Department of Homeland Security (DHS) place direct constraints on the designer in the interests of protecting health, safety, and security. Several aspects of the CPSC regulations have far-reaching influence on product design. Although the intended purpose of a product normally is quite clear, the unintended uses of that product are not always obvious. Under the CPSC regulations, the designer has the obligation to foresee as many unintended uses as possible, then develop the design in such a way as to prevent hazardous use of the product in an unintended but foreseeable manner. When unintended use cannot be prevented by functional design, clear, complete, unambiguous warnings must be permanently attached to the product. In addition, the designer must be cognizant of all advertising material, owner's manuals, and operating instructions that relate to the product to ensure that the contents of the material are consistent with safe operating procedures and do not promise performance characteristics that are beyond the capability of the design.

An important design consideration is adequate attention to human factors engineering, which uses the sciences of biomechanics, ergonomics, and engineering psychology to assure that the design can be operated efficiently and safely by humans. It applies physiological and anthropometric data to such design features as visual and auditory display of instruments and control systems. It is also concerned with human muscle power and response times.